

Public Rules, Executed Behavior, and Welfare in a Language-Model Economy

Anonymous Author(s)

Submission Id: TBD

Abstract

Public natural-language rules for language-model agents are usually judged by whether agents perform the action the rule names. We test that practice in the Verification Budget Economy, an eight-agent random-matching economy with scarce verification checks. A public announcement naming Easy-to-Hard gifts raises welfare by 2.06 points per account over a no-recommendation announcement (14/14 paired seeds), yet an exact accounting identity attributes the gain to Hard–Hard check swaps—mutual cooperation in an unnamed stage game that is a Prisoner’s Dilemma—net of dominated Hard-to-Easy giving the same text induces. Under a byte-identical gift announcement, blocking Hard–Hard transfers after the model responds lowers welfare by 3.21 (12/12 seeds). Replaying the frozen traces through the engine, we compare first decisions at states that are byte-identical except for the announcement: the named gift is engaged but collapses after one meeting, whereas cross-role Hard–Hard giving is raised directly by the text, including by a harmful Easy–Easy directive obeyed in 113 of 114 meetings. Explicit exclusion clauses bound the response; positive role predicates do not.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Agent / discrete models*; *Natural language processing*.

Keywords

language-model agents, multi-agent systems, public rules, norms, causal mechanisms, welfare, agent-based simulation

ACM Reference Format:

Anonymous Author(s). 2027. Public Rules, Executed Behavior, and Welfare in a Language-Model Economy. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027), Hanoi, Vietnam, 3–7 May 2027*, IFAAMAS, 8 pages.

1 Introduction

Teams, markets, and simulations of language-model agents increasingly coordinate through public rules written in natural language. The usual check of such a rule is whether agents perform the action it names and whether an aggregate outcome improves. That check conflates four objects: the proposals agents make, the transactions the environment executes from pairs of proposals, the state those transactions produce, and the welfare that results. A rule can be

followed in its named relation and still owe its effect to a relation it never mentions.

We study this gap in the Verification Budget Economy (VBE), where Hard agents need verification checks, Easy agents have spare ones, and the engine executes only compatible pairs of proposals. Three prospectively frozen studies with a hosted model supply the causal comparisons; a zero-call reanalysis of their frozen traces supplies the accounting and the fixed-state comparisons. The results form one chain:

- *Welfare comes from an unnamed relation.* The engine implies that a Hard–Hard meeting is a strict Prisoner’s Dilemma and that giving in an Easy–Hard meeting is individually dominated for both roles. A gift announcement naming Easy-to-Hard gifts raises welfare, and an exact accounting identity attributes the gain to Hard–Hard swaps net of wasteful Hard-to-Easy gifts. Blocking Hard–Hard transfers under the identical announcement removes the gain (§4.1).
- *The text acts on unnamed roles directly.* Because each prompt is a function of reconstructable state, we can find first decisions taken from states that are byte-identical across arms except for the announcement. There, the named Easy-to-Hard gift is engaged, and Hard–Hard giving is raised by the gift text and by a harmful Easy–Easy directive. Explicit exclusion clauses bound the response; positive role predicates do not (§4.2).
- *Closed-loop dynamics separate the two.* The named gift collapses after an Easy agent’s first meeting, whereas Hard–Hard cooperation and harmful Easy–Easy compliance persist for the whole run (§4.3).

The contribution is an identification design and the result it produces, not a reminder that intention differs from execution. Execution-layer intervention under a fixed prompt, exact transaction accounting, fixed-state comparison from frozen logs, and an engaged harmful control together rule out the intuitive explanation—that the rule works because agents do what it says—and locate both the welfare channel and the failure modes. The claim concerns a hosted model population in one economy, not private belief, common knowledge, or a self-enforcing institution.

2 Environment

2.1 Rules and state

VBE has $n = 8$ persistent accounts and $T = 24$ rounds; the supplement lists every parameter. All accounts start with score 0, and four random accounts start with one *mark* each. Marks have no direct or terminal value and persist across rounds. Each round proceeds as follows.



This work is licensed under a Creative Commons Attribution International 4.0 License.

Table 1: Expected direct payoff change of one executed transfer relative to no transfer, derived from the engine. A sale moves the same payoffs as a gift in the same direction because marks are valueless.

Executed transfer	Giver	Receiver	Total
H–H swap	+1.83	+1.83	+3.66
E–H swap (E is “giver”)	−0.50	+1.83	+1.33
E–E swap	−0.50	−0.50	−1.00
One-way E→H	−0.50	+2.33	+1.83
One-way H→H	−0.96	+2.33	+1.37
One-way H→E	−0.96	0.00	−0.96
One-way E→E	−0.50	0.00	−0.50

- (1) *Roles and endowments.* Exactly four accounts are drawn Hard and four Easy. Every account receives $B = 1$ check. An Easy account is paid $R = 3$ immediately.
- (2) *Matching.* Accounts are shuffled into four pairs, and each pair meets independently with probability $q = 0.4$, so an account has at most one meeting per round and roles are independent of pairing.
- (3) *Proposals.* Each party to a meeting simultaneously proposes giveCheck, giveChits (0 or 1 marks), and requireChit.
- (4) *Resolution.* In priority order: if both give and neither requires a mark, the checks are *swapped*; if one gives while requiring a mark and the other offers one, a *sale* executes; if one gives without requiring a mark, it is a one-way *gift*; otherwise nothing moves. A received check is a verification source; it is not added to the recipient’s inventory.
- (5) *Settlement.* A Hard account that received a check uses it and succeeds with probability $p_P = 0.93$; otherwise it uses its own check and succeeds with probability $p_H = 0.32$. Success pays R . Every check still in inventory pays $v = 0.5$, and checks expire.

Scores, marks, and each account’s last $K = 4$ meetings persist. The prompt shows the round, the agent’s role, checks, marks, and score, the partner’s role, checks, and marks, and the agent’s own last four meetings; it never shows the partner’s score or history. One payoff draw per account per round is pre-generated from the seed, so arms sharing a seed share roles, pairings, and payoff draws.

2.2 Transaction payoffs and stage games

Table 1 lists the expected change for every transfer the engine can execute. Each entry is checked against the engine by integrating its settlement over the payoff-draw strata, and an exhaustive test covers all 4,096 local meeting states (§3.1). Two rows matter most. An E–H swap is worth 1.33, not 1.83: the Hard agent gives away a check it would otherwise salvage, and the Easy agent cannot use the check it receives. A one-way H→E gift destroys 0.96.

The identities imply the stage games. An H–H meeting is a strict Prisoner’s Dilemma (Table 2): keeping is dominant, and mutual giving maximizes welfare. In an E–H meeting, keeping is strictly dominant for both roles, and the welfare-maximizing outcome is a one-way E→H gift, which costs the Easy agent 0.5. Every welfare-improving transfer therefore requires an agent to give up private payoff. The population game is repeated under random rematching

Table 2: Expected round payoffs (row, column) in a Hard–Hard meeting implied by Table 1. Keeping is strictly dominant and mutual giving maximizes welfare: a strict Prisoner’s Dilemma.

	Partner keeps	Partner gives
Keep	(0.96, 0.96)	(3.29, 0.00)
Give	(0.00, 3.29)	(2.79, 2.79)

with private four-meeting histories, where cooperation can in principle be sustained by community enforcement [12, 21]; our design does not identify such a mechanism.

Two model-free benchmarks run on the same seeds, schedules, and draws. The *ex-ante first best* maximizes expected welfare meeting by meeting over the engine’s action space without using realized draws: an H–H swap, a one-way E→H gift, and no transfer otherwise. It is a planner benchmark, not an equilibrium. The *ex-post first best* uses realized draws and bounds attainable welfare. *Never-transfer* sets the baseline.

3 Studies and Inference

All model studies use deepseek-flash in one provider-labeled serving era at temperature zero. Every protocol, seed list, prompt, contrast, and gate was hash-frozen locally before its first target call; none was externally registered. Each study is bracketed by a 14-call provider-health check before the first and after the last target call, with external controls, dominance anchors, entropy, returned-model, catalog, and fingerprint gates; all brackets reported here are healthy and identity-consistent. The supplement records the anomalous earlier service window that motivated the bracket.

W-CO (discovery). Gift versus money announcements on 18 fresh paired seeds. Its welfare contrast was prospective; its planned gift-for-sale substitution gate failed, and the H–H channel was found afterwards. We use it only as discovery and as a third replication of the fixed-state results.

W-RG (execution intervention). Four arms on 12 fresh paired seeds: neutral; gift; gift with H–H check giving blocked; gift with unconditional E→H give proposals blocked. The three gift arms show the model the same announcement. After the model responds, a filter sets only giveCheck to false on eligible proposals and leaves the other fields unchanged; the engine then executes as usual and later prompts reflect the changed history. W-RG estimates the dynamic total effect of changing the execution contract under a fixed announcement.

W-SGB (package boundary). Seven announcements on 14 fresh paired seeds under unmodified execution (Table 3). The harmful E–E directive names a transfer that destroys 1.0 per execution and carries its own frozen engagement gate. Packages other than the three historical anchors are not length-matched, so contrasts identify whole packages.

Inference. Let $Y_s(d, e)$ be the seed- s population outcome under announcement d and execution contract e ; the seed fixes initial state, roles, pairings, and payoff draws. W-RG and W-SGB estimate,

Table 3: W-SGB announcement packages. Every package also states that it is a public statement rather than an engine rule, may be followed or ignored, and that agents should maximize their own score. The supplement reproduces every package verbatim.

Arm	Distinguishing clause (abridged)	Named relation; explicit exclusion
Neutral	“announces no exchange recommendation”	none
Gift exact	“an Easy agent gives a leftover check to a Hard agent”	$E \rightarrow H$ gift
Easy only	“only an Easy agent gives ... A Hard agent does not give”	$E \rightarrow H$ gift; excludes Hard actors
Any holder	“any agent who holds a leftover check gives it to a Hard agent”	$\{E, H\} \rightarrow H$ gift
Hard partner	“an agent gives ... only when the current partner is Hard ... do not give ... when the current partner is Easy”	$\{E, H\} \rightarrow H$ gift; excludes Easy partners
Money exact	“a mark is accepted as payment ... a Hard agent who holds a mark may buy”	$E \rightarrow H$ sale
Harmful E-E	“when two Easy agents meet ... each gives that check to the other”	$E \leftrightarrow E$ swap

respectively,

$$\tau_{RG} = \frac{1}{12} \sum_s [Y_s(d_{\text{gift}}, e_{\text{std}}) - Y_s(d_{\text{gift}}, e_{\text{-HH}})],$$

$$\tau_{SGB} = \frac{1}{14} \sum_s [Y_s(d_1, e_{\text{std}}) - Y_s(d_0, e_{\text{std}})].$$

Both include downstream feedback through holdings and histories, so the inferential unit is the paired seed-level trajectory, not a meeting or account, and neither identifies a natural indirect effect [5, 19]. We report percentile bootstrap intervals (20,000 resamples of seed differences; pointwise, not simultaneous) and one-sided exact sign-flip p -values for $\Delta > 0$ under a zero-centered sign-symmetry null. W-SGB applies Holm correction at 0.025 within its eight H-H and eight welfare contrasts. A frozen gate passes when the mean reaches the minimum relevant effect size (MRES: 0.25 for swap rate, 0.5 for welfare) and the adjusted $p \leq 0.025$. Passing a gate shows a large estimate and a nonzero effect; it does not show $\Delta > MRES$. We therefore add supplementary exact tests of $H_0: \Delta \leq MRES$ and leave-one-seed-out ranges. With 12 or 14 seeds the smallest attainable one-sided p is 2^{-12} or 2^{-14} , reached whenever every seed agrees in sign, so we report sign counts throughout.

3.1 Zero-call reanalysis of frozen traces

The reanalysis adds no model call and changes no frozen artifact. We replay every frozen run through the frozen engine by feeding back its recorded proposals. Because roles, pairings, and draws come from the seed alone, the replay reproduces all 182 runs exactly (every final score and every meeting resolution), recovering the exact prompt behind each of the 14,010 decisions, a count equal to the three studies’ frozen call totals. This supports three analyses. *Accounting*: each run’s welfare equals its never-transfer welfare plus the realized contribution of each executed transfer, computed from the regenerated payoff draws; the identity holds exactly in all 182 runs. *Fixed states*: at an agent’s first meeting, its prompt carries no treatment-dependent history, and the reconstructed prompts are byte-identical across arms once the announcement is removed (verified for every first meeting); comparing arms there isolates the announcement’s effect on that decision. *Engine validation*: an exhaustive test enumerates every role pair, inventory, and proposal combination (4,096 cases) and checks resource conservation, the filters, and settlement against Table 1. These analyses are post hoc and descriptive with respect to the frozen studies.

4 Results

4.1 Welfare comes from an unnamed cooperation channel

The gift announcement raises welfare. In W-SGB it exceeds neutral by 2.0625 points per account ([1.429, 2.813], 14/14 seeds), raising the H-H swap rate by 0.607; W-CO’s gift-minus-money contrast (+2.333, 17/18 seeds) and W-RG’s gift-minus-neutral contrast (+1.682, 9/12 seeds) agree in sign (Table 5).

Where the gain comes from. The named transfer is almost absent from the realized trace. In W-SGB’s gift arm, 105 of 125 H-H meetings end in swaps (27 under neutral), while only 9 of 304 E-H meetings end in a one-way $E \rightarrow H$ gift. Figure 1B decomposes the welfare difference exactly: H-H swaps contribute +2.20 per account and E-H swaps +0.36, while 99 one-way $H \rightarrow E$ gifts (3 under neutral) cost 0.62. W-RG gives the same pattern (+2.41 from H-H swaps, -1.13 from $H \rightarrow E$ gifts). The announcement therefore buys welfare through cooperation in the unnamed Prisoner’s Dilemma and pays for part of it with a dominated, welfare-destroying transfer the same text induces. Reciprocity also costs welfare in the named relation: in W-RG’s gift arm, 25 of 26 feasible $E \rightarrow H$ gift intentions met an unconditional Hard give and resolved as E-H swaps, each worth 0.5 less than the one-way gift (12.5 points in total). Joint proposals show where these transfers come from: in the gift arm’s E-H meetings, only the Easy agent gave in 9, only the Hard agent in 99, both in 28, and neither in 168. Table 4 places every package against model-free benchmarks on the same seeds. The gift package captures 35% of the ex-ante feasible gain; the Hard-partner package captures the most, 38%, because it keeps H-H swaps (115) while nearly eliminating $H \rightarrow E$ gifts (18). Mutual H-H giving is not simply two independent proposals: under neutral it occurs in 27 of 125 meetings, against about 8 expected from the marginal give rate, whereas under gift it matches independence (105 versus 104).

Execution intervention. W-RG holds the announcement fixed and removes the H-H channel after the model responds. Welfare falls by 3.208 ([2.469, 3.891], 12/12 seeds, Figure 1A), from 55.30 to 52.09—below the same seeds’ never-transfer benchmark (52.53) in 10 of 12 seeds. Without H-H circulation, the remaining transfers induced by the gift announcement are net negative. Removing transfers worth 3.66 each will lower welfare; the result is informative because the removed channel is the one the announcement does not name, and because what remains of the announcement’s effect is wasteful. The $E \rightarrow H$ filter changed little: of its 20 blocked proposals, 16 would have produced E-H swaps and instead executed as one-way $H \rightarrow E$

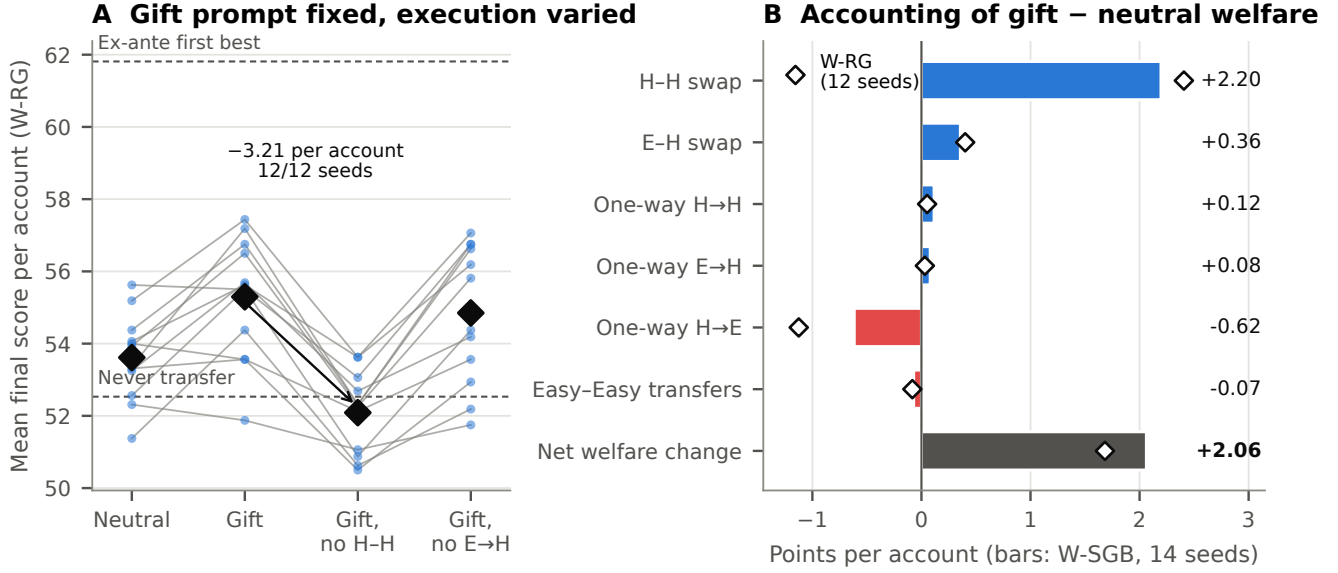


Figure 1: (A) W-RG: seed-level welfare (lines connect a seed’s four paired runs). Blocking H-H transfers under the identical gift announcement removes the gift’s gain. Dashed lines are model-free benchmarks on the same seeds. (B) Exact accounting of the gift-minus-neutral welfare difference by executed transfer type (W-SGB bars; W-RG diamonds). This is an accounting identity, not a mediation estimate: arms differ in which transfers occur.

Table 4: Absolute welfare and executed transfers by W-SGB package (14 seeds, same schedules and payoff draws). Gain is welfare minus the never-transfer benchmark; share is gain divided by the ex-ante first best’s gain. Transfer columns count executed meetings out of 125 H-H, 304 E-H, and 114 E-E meetings per arm.

Package or benchmark	Welfare	Gain	Share	H-H swaps	E-H swaps	One-way E→H	One-way H→E	E-E swaps
Neutral	55.28	1.04	0.12	27	0	0	3	0
Gift exact	57.34	3.10	0.35	105	28	9	99	2
Easy only	55.23	0.99	0.11	29	0	0	16	1
Any holder	57.25	3.00	0.34	111	28	7	95	0
Hard partner	57.68	3.44	0.38	115	1	4	18	0
Money exact	55.07	0.83	0.09	21	0	0	0	0
Harmful E-E	55.46	1.22	0.14	75	1	3	35	113
Never transfer	54.24	0	0	0	0	0	0	0
Always give	60.81	6.57	0.73	125	304	0	0	114
Ex-ante first best	63.19	8.95	1	125	0	304	0	0
Ex-post first best (uses realized draws)	63.26	9.02	1.01	—	—	—	—	—

gifts, and 4 would have produced one-way E→H gifts. The regime contrast “E→H-blocked minus H-H-blocked” (+2.760, 12/12) therefore compares keeping H-H circulation with keeping the little that the E→H path carried.

W-RG has no neutral-with-H-H-blocked arm, so it does not identify how much of the gift-versus-neutral effect depends on the H-H path; the accounting identity above is consistent with nearly all of it but is not a causal decomposition. The supplement gives a frozen-ready protocol for the announcement-by-execution factorial that would.

The stricter supplementary test confirms every passing frozen gate, with one exception: the harmful directive’s H-H effect rejects $\Delta \leq 0.25$ before correction ($p = 0.0089$) but not after Holm (0.027). Every passing contrast keeps its mean above MRES when any single seed is dropped.

4.2 The text acts on unnamed roles directly

Aggregate role-conditional proposal rates mix two sources: the announcement changes what an agent does in a given state, and it changes which states agents reach. First decisions remove the second source. Figure 2A shows, for every W-SGB package, the unconditional-give rate at each agent’s first meeting, where the prompt differs across arms only in the announcement (18 H→H, 35 H→E, 39 E→H, and 20 E→E decisions per arm).

The named relation is engaged. Under gift exact, 17 of 39 Easy agents give to a Hard partner at their first meeting and none do under neutral (17 discordant pairs in one direction, 0 in the other). The same holds in W-CO (26 of 41) and W-RG (16 of 31). The low closed-loop rate of named giving (0.123 above neutral in W-SGB) therefore does not reflect a failure to engage the named relation;

Table 5: Prospective paired contrasts. CIs are pointwise percentile bootstrap intervals over seeds; $+/-/0$ counts seeds by sign. p is the one-sided exact sign-flip p for $\Delta > 0$, Holm-adjusted within each W-SGB endpoint family. p_{MRES} tests $H_0: \Delta \leq MRES$ (supplementary; Holm within each W-SGB family). Frozen gates require mean $\geq MRES$ and $p \leq 0.025$.

W-RG (12 seeds)		Δ	95% CI	$+/-/0$	p	p_{MRES}	Gate
Gift – neutral, welfare		1.682	[0.854, 2.463]	9/3/0	.0034	.011	pass
Gift – H-H-blocked, welfare		3.208	[2.469, 3.891]	12/0/0	.00024	.00024	pass
E→H-blocked – H-H-blocked, welfare		2.760	[2.094, 3.412]	12/0/0	.00024	.00024	pass
Gift – H-H-blocked, H-H swap rate		0.753	[0.633, 0.862]	12/0/0	.00024	.0020	pass

H-H swap rate (MRES 0.25)						Welfare (MRES 0.5)				
W-SGB (14 seeds)	Δ	95% CI	$+/-/0$	p	p_{MRES}	Δ	95% CI	$+/-/0$	p	p_{MRES}
Gift – neutral	.607	[.510, .714]	14/0/0	.00049	.00049	2.063	[1.429, 2.813]	14/0/0	.00049	.00085
Gift – money	.663	[.568, .755]	14/0/0	.00049	.00049	2.268	[1.670, 2.960]	14/0/0	.00049	.00049
Gift – harmful E-E	.229 [†]	[.105, .351]	10/3/1	.0054	1.0	1.875	[1.348, 2.433]	14/0/0	.00049	.0011
Harmful E-E – neutral	.378	[.287, .474]	14/0/0	.00049	.027	.188	[-.339, .710]	9/5/0	.51	1.0
Gift – Easy only	.592	[.482, .697]	14/0/0	.00049	.00049	2.112	[1.464, 2.813]	14/0/0	.00049	.0011
Any holder – neutral	.658	[.549, .771]	14/0/0	.00049	.00049	1.969	[1.295, 2.696]	14/0/0	.00049	.0011
Hard partner – neutral	.703	[.596, .810]	14/0/0	.00049	.00049	2.406	[1.759, 3.107]	14/0/0	.00049	.0011
Money – neutral	-.056	[-.126, -.001]	1/5/8	.97	1.0	-.205	[-.339, -.071]	1/9/4	.996	1.0

[†]Significant but below the 0.25 MRES, so its frozen gate fails. Every other row with mean $\geq MRES$ and $p \leq 0.025$ passes its frozen gate.

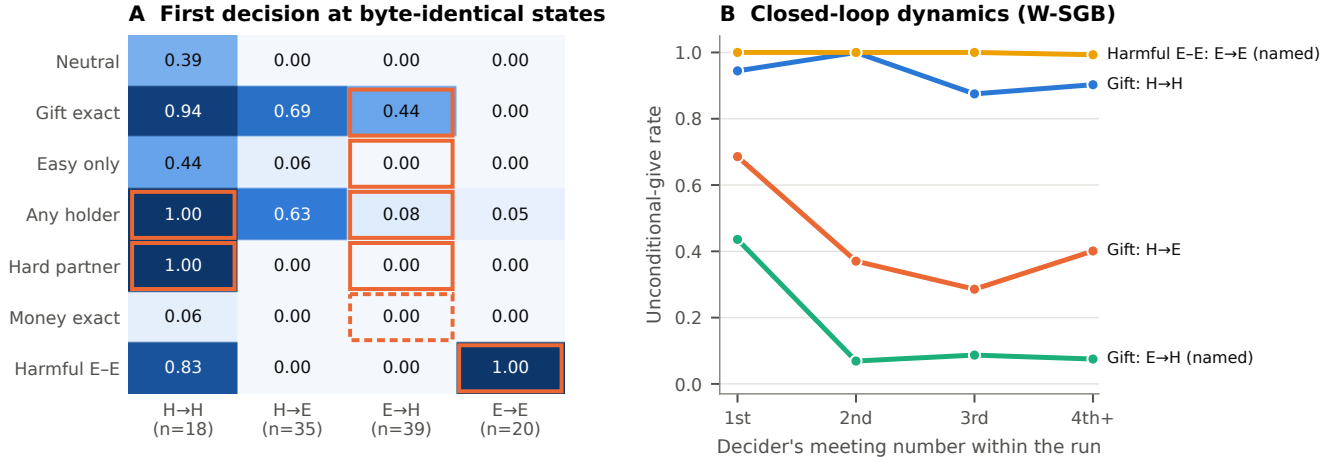


Figure 2: (A) Unconditional-give rate at each agent's first meeting, where prompts are byte-identical across arms except for the announcement (W-SGB). Orange outlines mark the relation each package names (dashed: it names a sale, not a gift). (B) Give rate by the decider's meeting number within the run; later meetings condition on realized history, so this panel is descriptive.

§4.3 shows it reflects decay. A post-hoc engagement screen that compared closed-loop named rates with 0.25 would call the gift package unengaged and the any-holder and Hard-partner packages engaged; at first decisions the classification reverses, and in closed loop it flips again between thresholds of 0.10 and 0.15 (supplement). We therefore treat that screen as descriptive only.

Cross-role giving is a direct text effect. At first decisions, Hard agents meeting a Hard partner give at rate 0.39 under neutral, 0.94 under the gift package, and 0.83 under the harmful E-E directive, which names only Easy agents. All disagreements with neutral run one way (10/0 and 8/0 discordant decisions). The harmful directive's H-H spillover is therefore not a by-product of the trajectories it creates. Under the gift package, Hard agents also give to Easy partners (0.69), outside both the named actor and the named beneficiary.

Exclusion clauses bind; positive predicates do not. Packages that *exclude* a relation suppress it: “do not give when the partner is Easy” reduces first-decision H→E giving to 0.00 (from 0.69 under gift exact), and “a Hard agent does not give” leaves H→H giving at the neutral level (0.44 versus 0.39). Packages that only *name* a relation do not confine the response to it: any holder names Hard beneficiaries, yet Hard agents give to Easy partners at rate 0.63. Naming the actor matters for Easy agents: they give at first decisions when the package names Easy actors (0.44), not when it names “any agent” (0.08) or “an agent” (0.00). The Easy-only package suppresses its own named relation (0/39), so it cannot isolate actor exclusivity; the money package lowers first-decision H-H giving (0.06, six discordant decisions all favoring neutral).

Noise floor. Temperature zero does not make the endpoint deterministic [6]. Across W-RG’s three gift arms, decisions with byte-identical full prompts returned identical proposals in 87–90% of 877 comparisons, and disagreements were symmetric (10 versus 16, 16 versus 15, and 13 versus 11 give-direction splits). The one-directional splits above (17/0, 10/0, 8/0) lie far outside that noise.

4.3 Closed-loop dynamics

Figure 2B follows the same decisions over each agent’s successive meetings. The named E→H gift collapses after an Easy agent’s first meeting in all three studies (0.63 → 0.02 in W-CO, 0.52 → 0.00 in W-RG, 0.44 → 0.07 in W-SGB). H→E giving declines more slowly (0.69 to 0.29–0.40). H→H giving stays near 0.9 throughout, so mutual cooperation in the Prisoner’s Dilemma persists without any enforcement. Harmful E–E compliance does not decay: E–E giving remains 0.99 at agents’ fourth and later meetings, although every execution costs each Easy agent 0.5 of salvage. These trends condition on realized history and are descriptive; they are consistent with agents abandoning unreciprocated sacrifices while maintaining reciprocated ones, but the design does not identify that mechanism.

4.4 A harmful directive and an unexecuted money package

The harmful directive adds no rule, reward, monitoring, or sanction and says it may be ignored, yet it executes 113 E–E swaps in 114 opportunities (frozen engagement contrast 0.982, [0.946, 1.000], adjusted $p = 0.00006$). Its net welfare effect is +0.19 ([−0.34, 0.71]), because H–H swaps it induces (+1.23 per account) offset the 1.01 it destroys through named E–E swaps. Behavioral spread across roles and welfare are thus separate quantities. Whether agents accept the salvage loss knowingly or fail to represent it remains open; the supplement specifies a payoff-salience replay that separates the two.

No arm of any study executed a mark sale. The funnel locates the failure. Under money exact, the Hard partner held a mark in 140 of 304 E–H meetings, the Easy seller proposed a mark-contingent sale in all 140 (136 of 140 under neutral), and the Hard buyer offered a mark in none. Across all 13 arms of the three studies, Hard agents offered a mark in 0 of 3,443 decisions taken while holding one—including under a money announcement that tells mark holders they may buy. The money package therefore tests nothing about priced semantics; its failure is buyer non-payment.

5 Discussion

What is identified. Within the gift announcement, the H–H execution channel carries the welfare gain (W-RG), and the accounting identity attributes the gift-versus-neutral gain to H–H swaps net of H→E waste. At fixed states the announcement directly raises giving by unnamed actors toward unnamed beneficiaries, and only explicit exclusions bound that response. Over time, unreciprocated named giving decays while reciprocated cooperation and harmful reciprocal compliance persist. The best description of the object is a *distributed policy program*: public text changes a population’s mapping from role and state to proposals, and the engine determines which proposals become transitions. It is not an institution in the stronger sense of common belief, legitimacy, consent, or

self-enforcement [31, 39], and every agent receives the announcement directly, so cross-role responses are population responses, not peer-to-peer contagion.

Design implications. Version natural-language rules like code and log the proposal, execution contract, and transition behind every outcome. Evaluate a rule on the relations it does not name, not only on the one it names. State exclusions explicitly, because in this population a positive scope predicate did not confine behavior and an exclusion did. Measure welfare against model-free benchmarks on the same schedules, because named compliance, cross-role spread, and welfare moved independently: the most compliant package destroyed value in its named relation, and the most effective package was not the one that named the welfare-maximizing transfer. Finally, when prompts are a function of logged state, frozen traces support exact replay, fixed-state comparison, and exact accounting at no inference cost.

6 Limitations

All model evidence comes from one hosted model in one provider-labeled serving era; health brackets bound the service interval but do not establish immutable weights or cross-model generality. The only cross-family evidence is three historical grok-4.5 seeds, too few to count as replication. Packages are not length- or token-matched. Fixed-state comparisons cover first decisions only; a cross-announcement replay of later states with identical full histories requires new model calls. W-RG lacks the neutral-with-H–H-blocked arm needed to identify how much of the gift effect depends on H–H circulation. The economy fixes price, supply, horizon, and technology, and mean score is its only welfare criterion. The H–H endpoint was selected after W-CO’s planned mechanism failed; later studies test that selected phenotype on fresh seeds. The re-analysis in §3.1 is post hoc. The supplement provides frozen-ready protocols for the factorial, the full-history replay, a payoff-salience test of the harmful directive, a buyer-side money diagnostic, and a second-model replication.

7 Related Work

Norms and institutions. Institutional grammar and normative multi-agent systems formalize role-indexed obligations, permissions, monitoring, and sanctions [8, 11, 14, 17, 20, 22]; social-law synthesis and norm-emergence work study top-down constraints and bottom-up convergence [29, 37, 38, 40, 41]. These traditions specify what a rule means or authorizes. We ask what a natural-language rule executes when model policies interpret it, and find that its realized actor and beneficiary scope exceeds the written one unless an exclusion is stated. Convention research separates shared observation from common knowledge [9, 10, 25, 36]; our announcement is common exposure, and we infer no epistemic state.

Language-model populations. Generative agent-based models, simulated samples, and economic agents [2–4, 13, 18, 26, 27, 32, 44] and studies of coordination, cooperation, conventions, and legal or economic structure in LLM collectives [1, 15, 28, 34, 35, 43, 45] show what such populations can do. Concordia separates agents’

described actions from outcomes a game master resolves [42]; Gov-Sim pairs aggregate outcomes with communication ablations and capability probes [33]; Welfare Diplomacy scores general-sum welfare [30]. Our design adds three elements we have not found combined in this literature: execution-layer intervention under a fixed prompt, an exact transaction-level welfare identity, and comparison of decisions at states that are identical across treatments. Together they show that the named relation is not the welfare channel. The naming-game debate shows that collective behavior can reproduce training data [7, 16], so we claim no emergence. Search-theoretic money motivates the token environment [23, 24]; fixed supply, horizon, and prices preclude a monetary-equilibrium claim.

8 Conclusion

A public sentence changed a language-model population’s welfare, but not through the transaction it named. The gain came from cooperation in an unnamed Prisoner’s Dilemma, net of waste the same sentence induced; the text acted on unnamed roles directly; and only explicit exclusions bounded it. Evaluating such rules requires auditing named compliance, cross-role response, executed transitions, and welfare as separate quantities.

AI assistance and artifacts. OpenAI Codex (GPT-5-based agent; exact backend revision unavailable) and Anthropic Claude assisted design iteration, code, analysis, and editing; the authors are responsible for all content, and no AI system is an author. The anonymized supplement accompanying this submission contains the frozen protocols, manifests, traces, health brackets, engine, analyzers, the zero-call reanalysis, and the AI-use record required by the AAMAS policy.

References

- [1] Saaket Agashe, Yue Fan, Anthony Reyna, and Xin Eric Wang. 2025. LLM Coordination: Evaluating and Analyzing Multi-agent Coordination Abilities in Large Language Models. In *Findings of the Association for Computational Linguistics: NAACL 2025*. Association for Computational Linguistics, Albuquerque, New Mexico, 8053–8072. doi:10.18653/v1/2025.findings-naacl.448
- [2] Gati V. Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*. PMLR, 337–371.
- [3] Jacy Reese Anthis, Ryan Liu, Sean M. Richardson, Austin C. Kozlowski, Bernard Koch, Erik Brynjolfsson, James Evans, and Michael S. Bernstein. 2025. Position: LLM Social Simulations Are a Promising Research Method. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 81005–81034.
- [4] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis* 31, 3 (2023), 337–351. doi:10.1017/pan.2023.2
- [5] Peter M. Aronow and Cyrus Samii. 2017. Estimating Average Causal Effects under General Interference, with Application to a Social Network Experiment. *The Annals of Applied Statistics* 11, 4 (2017), 1912–1947. doi:10.1214/16-AOAS1005
- [6] Berk Atıl, Sarp Aykent, Alexa Chittams, Lisheng Fu, Rebecca J. Passonneau, Evan Radcliffe, Guru Rajan Rajagopal, Adam Sloan, Tomasz Tudrej, Ferhan Ture, Zhe Wu, Lixinyu Xu, and Breck Baldwin. 2025. Non-Determinism of “Deterministic” LLM System Settings in Hosted Environments. In *Proceedings of the 5th Workshop on Evaluation and Comparison of NLP Systems*. 135–148. <https://aclanthology.org/2025.evalnlp-1.12>
- [7] Christopher Barrie and Petter Törnberg. 2025. Emergent LLM Behaviors Are Observationally Equivalent to Data Leakage. *arXiv preprint arXiv:2505.23796* (2025).
- [8] Guido Boella and Leendert van der Torre. 2004. Regulative and Constitutive Norms in Normative Multiagent Systems. In *Principles of Knowledge Representation and Reasoning: Proceedings of the Ninth International Conference (KR 2004)*. AAAI Press, 255–265.
- [9] Damon Centola and Andrea Baronchelli. 2015. The Spontaneous Emergence of Conventions: An Experimental Study of Cultural Evolution. *Proceedings of the National Academy of Sciences* 112, 7 (2015), 1989–1994.
- [10] Damon Centola, Joshua Becker, Devon Brackbill, and Andrea Baronchelli. 2018. Experimental Evidence for Tipping Points in Social Convention. *Science* 360, 6393 (2018), 1116–1119.
- [11] Sue E. S. Crawford and Elinor Ostrom. 1995. A Grammar of Institutions. *American Political Science Review* 89, 3 (1995), 582–600. doi:10.2307/2082975
- [12] Glenn Ellison. 1994. Cooperation in the Prisoner’s Dilemma with Anonymous Random Matching. *The Review of Economic Studies* 61, 3 (1994), 567–588.
- [13] Joshua M. Epstein. 2006. *Generative Social Science: Studies in Agent-Based Computational Modeling*. Princeton University Press, Princeton, New Jersey.
- [14] Marc Esteva, Julian Padget, and Carles Sierra. 2002. Formalizing a Language for Institutions and Norms. In *Intelligent Agents VIII (LNCS, Vol. 2333)*. 348–366.
- [15] Ariel Flint Ashery, Luca Maria Aiello, and Andrea Baronchelli. 2025. Emergent Social Conventions and Collective Bias in LLM Populations. *Science Advances* 11, 20 (2025), eadu9368.
- [16] Ariel Flint Ashery, Luca Maria Aiello, and Andrea Baronchelli. 2025. Reply to “Emergent LLM Behaviors Are Observationally Equivalent to Data Leakage”. *arXiv preprint arXiv:2506.18600* (2025).
- [17] Nicoletta Fornara and Marco Colombetti. 2008. Specifying and Enforcing Norms in Artificial Institutions. In *Proceedings of AAMAS*. 1481–1484.
- [18] John J. Horton. 2023. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus? *arXiv preprint arXiv:2301.07543* (2023).
- [19] Michael G. Hudgens and M. Elizabeth Halloran. 2008. Toward Causal Inference With Interference. *J. Amer. Statist. Assoc.* 103, 482 (2008), 832–842. doi:10.1198/016214508000000292
- [20] Andrew J. I. Jones and Marek Sergot. 1996. A Formal Characterisation of Institutionalised Power. *Logic Journal of the IGPL* 4, 3 (1996), 427–443. doi:10.1093/jigpal/4.3.427
- [21] Michihiro Kandori. 1992. Social Norms and Community Enforcement. *The Review of Economic Studies* 59, 1 (1992), 63–80.
- [22] Thomas C. King, Tingting Li, Marina De Vos, Virginia Dignum, Catholijn M. Jonker, Julian Padget, and M. Birna van Riemsdijk. 2015. A Framework for Institutions Governing Institutions. In *Proceedings of AAMAS*. 473–481.
- [23] Nobuhiro Kiyotaki and Randall Wright. 1989. On Money as a Medium of Exchange. *Journal of Political Economy* 97, 4 (1989), 927–954.
- [24] Nobuhiro Kiyotaki and Randall Wright. 1993. A Search-Theoretic Approach to Monetary Economics. *American Economic Review* 83, 1 (1993), 63–77.
- [25] David Lewis. 1969. *Convention: A Philosophical Study*. Harvard University Press.
- [26] Nian Li, Chen Gao, Mingyu Li, Yong Li, and Qingmin Liao. 2024. EconAgent: Large Language Model-Empowered Agents for Simulating Macroeconomic Activities. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Bangkok, Thailand, 15523–15536. doi:10.18653/v1/2024.acl-long.829
- [27] Michael W. Macy and Robert Willer. 2002. From Factors to Actors: Computational Sociology and Agent-Based Modeling. *Annual Review of Sociology* 28 (2002), 143–166. doi:10.1146/annurev.soc.28.110601.141117
- [28] Hachem Madmoun and Salem Lahlou. 2026. Communication Enables Cooperation in LLM Agents: A Comparison with Curriculum-Based Approaches. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*. Association for Computational Linguistics, Rabat, Morocco, 307–321. doi:10.18653/v1/2026.eacl-short.23
- [29] Andreas Morris-Martin, Marina De Vos, and Julian Padget. 2019. Norm Emergence in Multiagent Systems: A Viewpoint Paper. *Autonomous Agents and Multi-Agent Systems* 33, 6 (2019), 706–749. doi:10.1007/s10458-019-09422-0
- [30] Gabriel Mukobi, Hannah Erlebach, Niklas Lauffer, Lewis Hammond, Alan Chan, and Jesse Clifton. 2023. Welfare Diplomacy: Benchmarking Language Model Cooperation. *arXiv preprint arXiv:2310.08901* (2023).
- [31] Elinor Ostrom. 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge. doi:10.1017/CBO9780511807763
- [32] Joon Sung Park, Joseph C. O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, 1–22. doi:10.1145/3586183.3606763
- [33] Giorgio Piatti, Zhijing Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. 2024. Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. *arXiv preprint arXiv:2404.16698* (2024).
- [34] Zhengting Qi, Huangyuan Su, Ao Qu, Chenyu Wang, Yu Yao, Han Zheng, Kushal Chattopadhyay, Guowei Xu, Zihan Wang, Weirui Ye, Vijay Janapa Reddi, Ju Li, Paul Pu Liang, Himabindu Lakkaraju, Sham Kakade, and Yilun Du. 2026. Economy of Minds: Emerging Multi-Agent Intelligence with Economic Interactions. *arXiv preprint arXiv:2606.02859* (2026).
- [35] Siyue Ren, Zhiyao Cui, Rui Song, Zhen Wang, and Shuyue Hu. 2024. Emergence of Social Norms in Generative Agent Societies: Principles and Architecture. *arXiv preprint arXiv:2403.08251* (2024).

- [36] Ariel Rubinstein. 1989. The Electronic Mail Game: Strategic Behavior under Almost Common Knowledge. *American Economic Review* 79, 3 (1989), 385–391.
- [37] Bastin Tony Roy Savarimuthu and Stephen Cranefield. 2011. Norm Creation, Spreading and Emergence: A Survey of Simulation Models of Norms in Multi-Agent Systems. *Multiagent and Grid Systems* 7, 1 (2011), 21–54. doi:10.3233/MGS-2011-0167
- [38] Bastin Tony Roy Savarimuthu, Maryam Purvis, Stephen Cranefield, and Martin Purvis. 2007. Mechanisms for Norm Emergence in Multiagent Societies. In *Proceedings of the 6th International Joint Conference on Autonomous Agents and Multiagent Systems*. ACM, 1–3. doi:10.1145/1329125.1329335
- [39] John R. Searle. 2005. What Is an Institution? *Journal of Institutional Economics* 1, 1 (2005), 1–22. doi:10.1017/S1744137405000020
- [40] Sandip Sen and Stéphane Airiau. 2007. Emergence of Norms Through Social Learning. In *Proceedings of the 20th International Joint Conference on Artificial Intelligence*. 1507–1512.
- [41] Yoav Shoham and Moshe Tennenholtz. 1992. On the Synthesis of Useful Social Laws for Artificial Agent Societies (Preliminary Report). In *Proceedings of the Tenth National Conference on Artificial Intelligence*. AAAI Press, 276–281.
- [42] Alexander Sasha Vezhnevets, John P. Agapiou, Avia Aharon, Ron Ziv, Jayd Matyas, Edgar A. Duénez-Guzmán, William A. Cunningham, Simon Osindero, Danny Karmon, and Joel Z. Leibo. 2023. Generative Agent-Based Modeling with Actions Grounded in Physical, Social, or Digital Space Using Concordia. *arXiv preprint arXiv:2312.03664* (2023).
- [43] Yiding Wang, Yuxuan Chen, Fanxu Meng, Xifan Chen, Xiaolei Yang, and Muhan Zhang. 2026. Law in Silico: Simulating Legal Society with LLM-Based Agents. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, San Diego, California, United States, 8061–8104. doi:10.18653/v1/2026.findings-acl.396
- [44] Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. 2024. SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents. In *The Twelfth International Conference on Learning Representations*.
- [45] Kunlun Zhu, Hongyi Du, Zhaochen Hong, Xiaocheng Yang, Shuyi Guo, Zhe Wang, Zhenhailong Wang, Cheng Qian, Xiangru Tang, Heng Ji, and Jiaxuan You. 2025. MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Vienna, Austria, 8580–8622. doi:10.18653/v1/2025.acl-long.421